

Saikat Das

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📍 Varanasi, India

EDUCATION

Indian Institute of Technology (BHU), Varanasi
Bachelor of Technology in Mining Engineering; CGPA: 8.39/10.0

Varanasi, India
July 2023 to May 2027

TECHNICAL SKILLS

Languages Go, Python, C++, Dart
Backend Microservices, REST APIs, gRPC, Kafka, FastAPI
Databases PostgreSQL, Redis, Apache Solr, Memgraph
DevOps & UI Docker, Linux, Git, Grafana k6, Flutter, React

EXPERIENCE

Software Engineering Intern Jun 2026 – Jul 2026
COPIOUS | Tripple Double Remote

- Built a **Flutter** split-screen web portal with auto-save and rule-based validation for FIBA scoresheet digitization.
- Engineered a **VLM** OCR pipeline via **OpenRouter**, A/B testing models to optimize accuracy vs. cost.
- Reduced hallucination and token costs via a 2-pass pipeline using dynamic prompts and **JSON** key aliasing.
- Cut operator cost by **80%** (\$0.52 → \$0.10) and entry time by **73%** (30m → 8m) via pre-filled entry.

Flutter Developer Intern Dec 2024 – Jan 2025
Vrajpath iTech LLP Remote

- Reduced API requests by **~30%** via **Riverpod** state caching, validated in **Dart DevTools**.
- Integrated **Razorpay** for secure payments and **Google Maps** for real-time geolocation.

PROJECTS

TradeOrders – Order Matching Engine | Go, Kafka, PostgreSQL, Redis, Docker | [GitHub](#) Oct 2025
Solved lock contention in high-volume trading via an event-driven architecture, scaling throughput 18x.

- Decoupled API from DB via **Kafka**; throughput rose from **1k req/s** to **18.8k req/s** (0 errors).
- Partitioned Kafka topics by instrument for lock-free, sequential matching per symbol.
- Grafana k6** load tested at **7,624 VUs**: **18.2k req/sec** sustained, **p95 231ms** on Intel i5-13500HX.

Bluppi – Shared Music Playback | Go, gRPC, PostgreSQL, Redis, Solr, Flutter | [Backend](#) [Frontend](#) May 2025
A real-time synchronization engine that fixes network lag across distributed devices.

- Uses **PTP-style** clock negotiation and **quorum-based** buffering to coordinate playback.
- Tracks online presence via a **gRPC** gateway: **51k** concurrent connections, 0 errors, **p99 connect 15ms**.
- Live ephemeral chat via **Redis Pub/Sub**, social features via **Memgraph**, and a 500GB catalog via **Solr**.

QQL – Approximate Query Processing Shell | Python, PostgreSQL | [GitHub](#) Aug 2025
A shell sitting above PostgreSQL that accelerates massive dataset aggregations 10x using bounded statistical error.

- Uses a pilot query to estimate sample sizes, supporting **COUNT**, **SUM**, **AVG** with configurable error bounds.
- Benchmarked: **10x speedup** at 5% error bound; **6x** at 1% error (vs exact query).
- Special Mention, **e6Data Hackathon 2024**.

SubPaper – Reddit Wallpapers App (Archived) | Flutter Apr 2024
A mobile application that aggregates and caches high-res wallpapers directly from Reddit.

- Published Reddit wallpaper app to Play Store. **300+ downloads**, **4.5/5 rating**.
- Implemented on-device caching for fast content delivery and integrated **AdMob** for monetization.

ACHIEVEMENTS

BNY Hackathon | 1st Place (Transaction Processing and Settlement) April 2026
Nutanix Hackathon | 2nd Place (Real-time Workflow Engine) 2025
LeetCode (Rating: **1681**, **234 Solved**) | **Codeforces** (Max Rating: **1282**, **150+ Solved**) Present
e6Data Hackathon | Special Mention for QQL System 2024